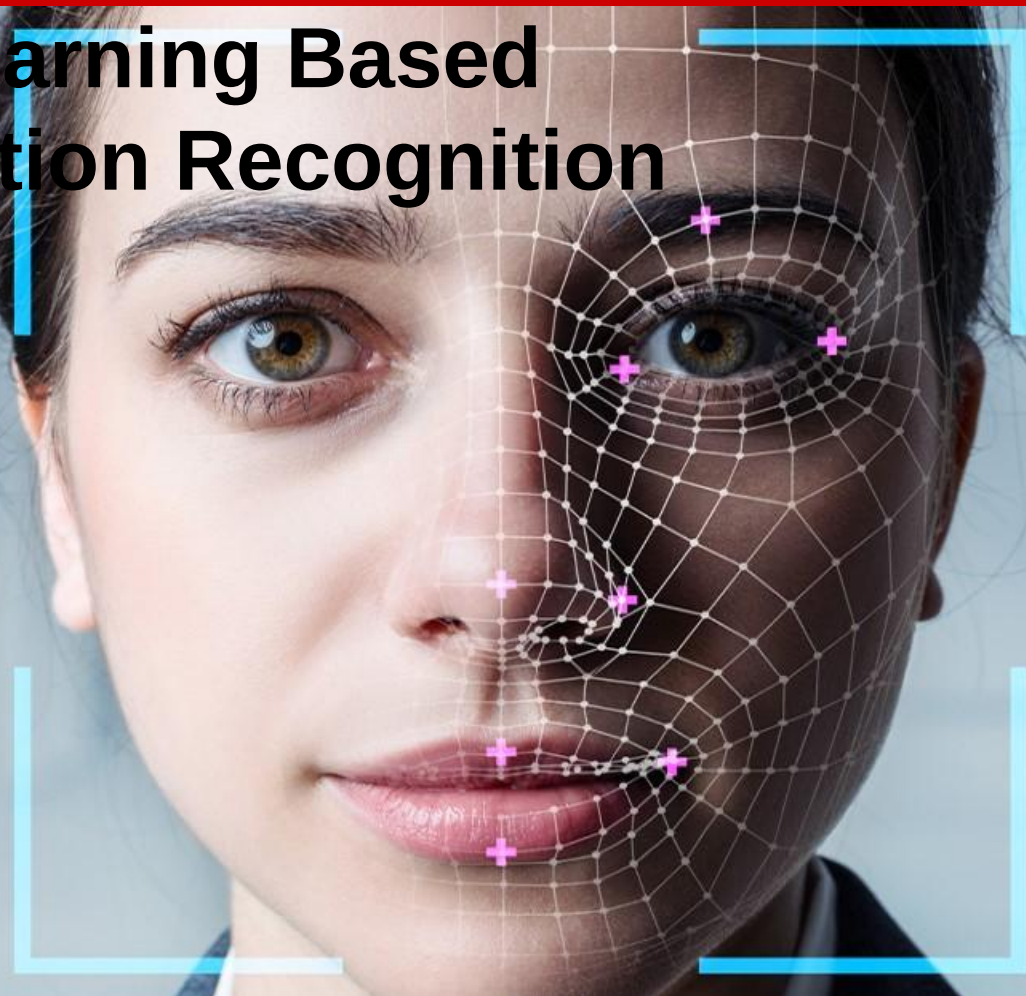


Deep Learning Based Facial Emotion Recognition

Reza Jahani



Problem Statement and Motivation

- Developing Intelligent Facial Recognition framework
- Facilitating human-computer interaction
- Enhancing customer service and security



Data and Input/Output

- Dataset: FER2013
- Grayscale 48x48 images
- Seven classes
 - Angry
 - Disgust
 - Fear
 - Happy
 - Neutral
 - Sad
 - Surprise

- Input: (1,48,48)
- Output: 7 Neurons

Class Imbalance Handled by adding augmented samples



Baseline Models

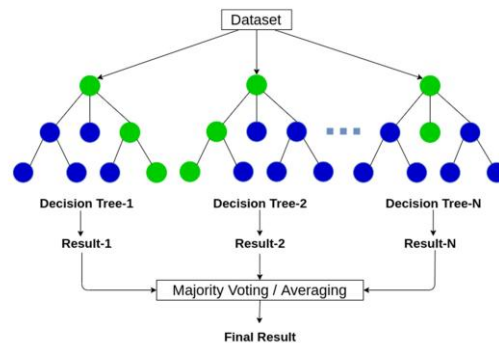
- Random Forest
 - Ensemble of decision trees used for classification
- MLP
 - Consisting of fully connected layers

```
Input Shape: torch.Size([2304])
```

Layer (type)	Output Shape	Param #
Linear-1	[-1, 512]	1,180,160
ReLU-2	[-1, 512]	0
Linear-3	[-1, 200]	102,600
ReLU-4	[-1, 200]	0
Linear-5	[-1, 100]	20,100
ReLU-6	[-1, 100]	0
Linear-7	[-1, 7]	707

Input Data Reshaped: (1,48,48) → (2304,)

Random Forest



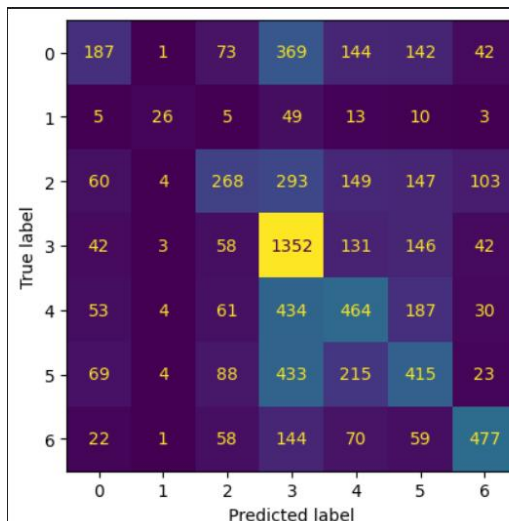
Random Forest Performance

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

Test Accuracy: 44.43%

Classification Report:

	precision	recall	f1-score	support
angry	0.43	0.20	0.27	958
disgust	0.60	0.23	0.34	111
fear	0.44	0.26	0.33	1024
happy	0.44	0.76	0.56	1774
neutral	0.39	0.38	0.38	1233
sad	0.38	0.33	0.35	1247
surprise	0.66	0.57	0.62	831
accuracy			0.44	7178
macro avg	0.48	0.39	0.41	7178
weighted avg	0.45	0.44	0.42	7178

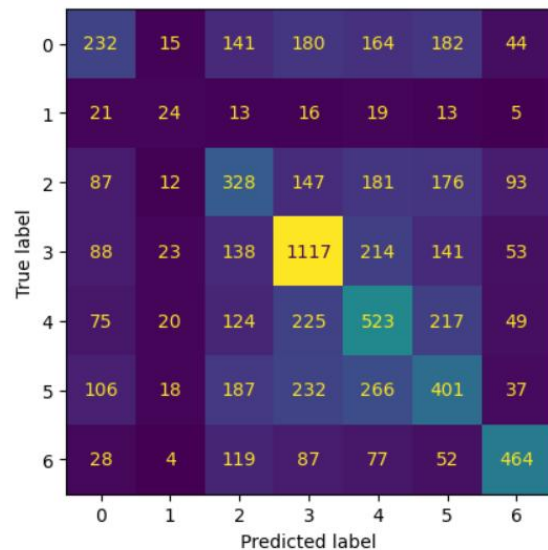


MLP Performance

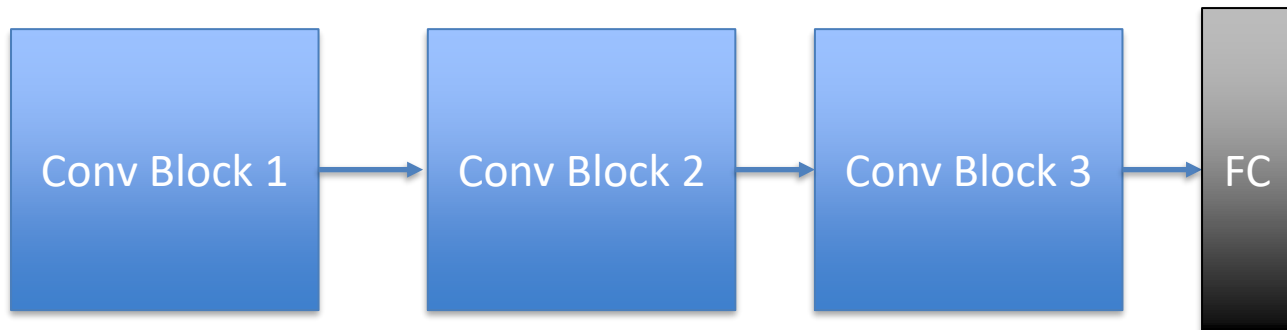
✓ Test Accuracy: 43.0343%

📊 Classification Report:

	precision	recall	f1-score	support
0	0.36	0.24	0.29	958
1	0.21	0.22	0.21	111
2	0.31	0.32	0.32	1024
3	0.56	0.63	0.59	1774
4	0.36	0.42	0.39	1233
5	0.34	0.32	0.33	1247
6	0.62	0.56	0.59	831
accuracy			0.43	7178
macro avg	0.40	0.39	0.39	7178
weighted avg	0.43	0.43	0.43	7178



Our Proposed CNN Architecture



```
self.block1 = nn.Sequential(  
    nn.Conv2d(1, 64, kernel_size=3, padding=1),  
    nn.BatchNorm2d(64),  
    nn.ReLU(),  
    nn.Conv2d(64, 64, kernel_size=3, padding=1),  
    nn.BatchNorm2d(64),  
    nn.ReLU(),  
    nn.MaxPool2d(kernel_size=2),  
    nn.Dropout(0.25)  
)
```

```
self.block2 = nn.Sequential(  
    nn.Conv2d(64, 128, kernel_size=3, padding=1),  
    nn.BatchNorm2d(128),  
    nn.ReLU(),  
    nn.Conv2d(128, 128, kernel_size=3, padding=1),  
    nn.BatchNorm2d(128),  
    nn.ReLU(),  
    nn.MaxPool2d(kernel_size=2),  
    nn.Dropout(0.25)  
)
```

```
self.block3 = nn.Sequential(  
    nn.Conv2d(128, 256, kernel_size=3, padding=1),  
    nn.BatchNorm2d(256),  
    nn.ReLU(),  
    nn.Conv2d(256, 256, kernel_size=3, padding=1),  
    nn.BatchNorm2d(256),  
    nn.ReLU(),  
    nn.MaxPool2d(kernel_size=2),  
    nn.Dropout(0.25)  
)
```

```
self.block4 = nn.Sequential(  
    nn.Linear(256 * 6 * 6, 512),  
    nn.BatchNorm1d(512),  
    nn.ReLU(),  
    nn.Dropout(0.5),  
    nn.Linear(512, num_classes)  
)
```

This architecture and training hyperparameters are based on simple grid search hyperparameter optimization.

CNN Performance

✓ Test Accuracy: 64.8971%

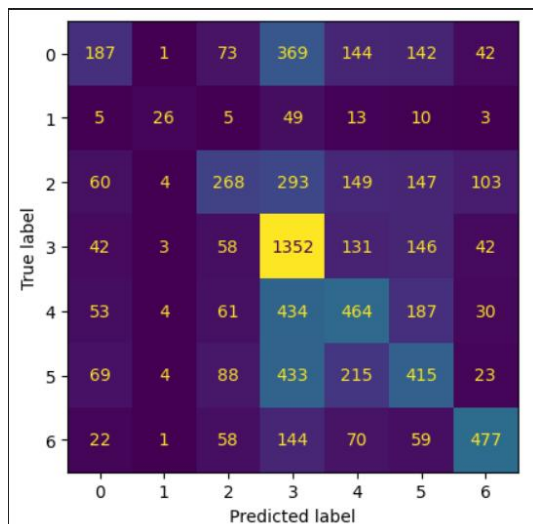
Classification Report:

	precision	recall	f1-score	support
0	0.56	0.54	0.55	958
1	0.72	0.57	0.64	111
2	0.50	0.48	0.49	1024
3	0.79	0.86	0.82	1774
4	0.57	0.60	0.59	1233
5	0.51	0.48	0.49	1247
6	0.79	0.77	0.78	831
accuracy			0.64	7178
macro avg	0.64	0.61	0.62	7178
weighted avg	0.63	0.64	0.63	7178

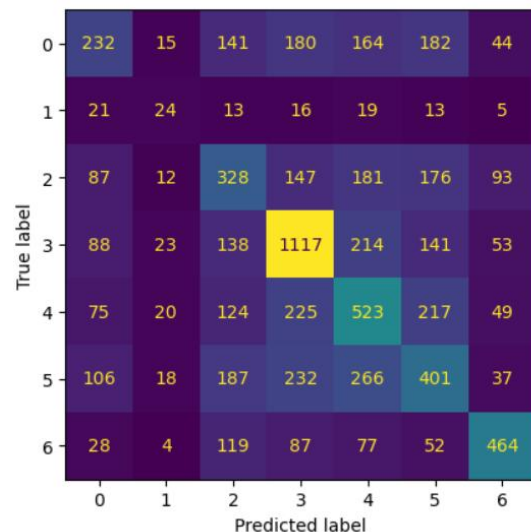
CNN Confusion Matrix

	0	1	2	3	4	5	6
0	517	11	108	71	98	134	19
1	24	63	12	2	3	6	1
2	107	3	488	52	104	185	85
3	31	2	43	1521	92	52	33
4	91	1	81	130	741	175	14
5	139	5	168	98	227	596	14
6	17	2	82	47	28	16	639
	0	1	2	3	4	5	6

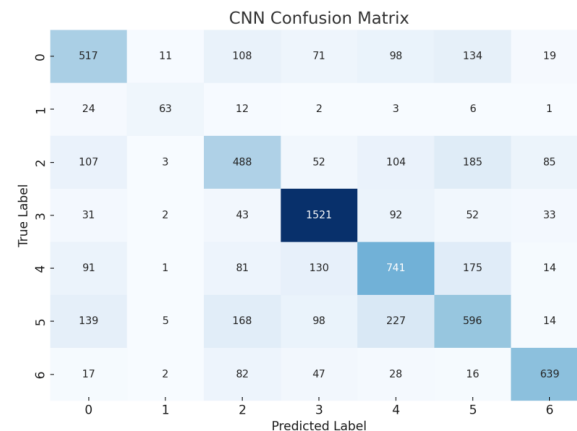
Performance Comparison



Random Forest
Test Accuracy: 44.43%



MLP
Test Accuracy: 43.03%



Our CNN:
Test Accuracy: 64.90%

Reference

- [1] <https://www.kaggle.com/datasets/msambare/fer2013>
- [2] <https://paperswithcode.com/dataset/fer2013>
- [3] Khaireddin, Yousif, and Zhuofa Chen. "Facial emotion recognition: State of the art performance on FER2013." arXiv preprint arXiv:2105.03588 (2021).

THANK YOU